The missing value heatmap

Where in the matrix the gaps are: which samples, which features, and whether the two are related.

AT A GLANCE

Where	Completeness → Missing Value Heatmap
Data levels	Protein, phosphosite, phosphoprotein. Peptide and precursor are not offered here.
Needs	An uploaded dataset and its annotation.
Shows	One cell per feature and sample: measured, or not.
Colours	The viridis scale over a yes/no matrix, so two colours only: pale yellow for a missing value, dark purple for a measured one. Fixed; there is no colour control on this page.
Table	No.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The missing value plot counts the gaps; this one places them. Its whole content is the pattern: a yellow column has missing measurements throughout, a yellow row is absent across samples, and a yellow block can show condition-associated missingness. These are observations, not diagnoses of sample failure or of a missing-data mechanism. A feature observed in only one condition may survive an at-least-one-condition filter; filling the other condition with low values can then strongly influence its fold change.

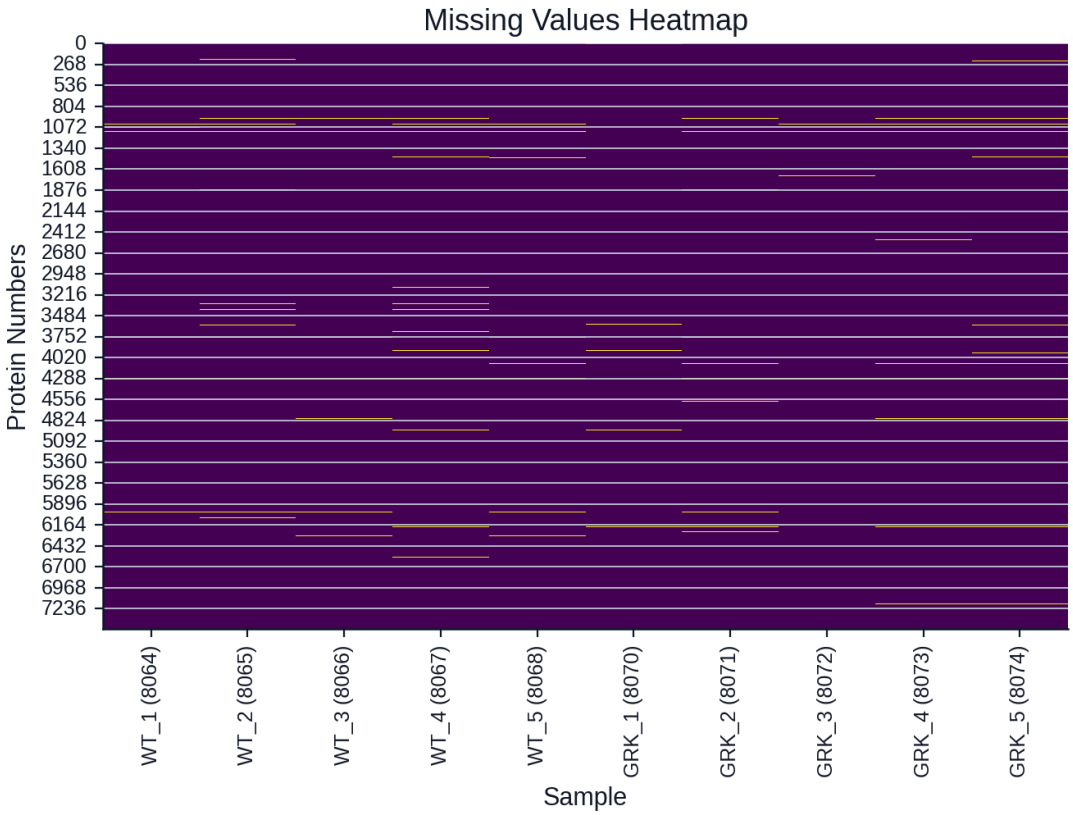


Figure 1.1. Protein level on the example project: ten columns, five WT then five GRK, against every protein of the matrix, with **Toggle IDs** on. Pale yellow is a missing value, dark purple a measured one; there is no colour bar, because the matrix has only two states. This dataset is almost complete, so the figure is nearly solid purple and the few gaps read as isolated horizontal lines — no column and no condition is paler than the rest.

Reading it

- One cell: one feature in one sample, yellow if that value is missing.
- A vertical yellow stripe: a sample missing a great many features. Check it against the coverage plot, where it is the short bar.
- A horizontal yellow band: features missing across most of the experiment, usually the low-abundance end of the dataset.
- A yellow block confined to the samples of one condition: missingness that is not random with respect to the comparison to be made.
- The label under each column: condition_n, numbered within the condition. With **Toggle IDs** on it becomes condition_n (id), where the id is the last number in the original sample name, or the whole name when it has no number.

> **EASY TO GET WRONG**
>
> Neither axis is sorted. The rows are in the order the matrix has them — the order of the uploaded file — and the columns are in the order of the annotation, grouped by condition. Nothing is clustered and nothing is reordered to make a pattern, which is what makes a visible block worth something; but it also means that a dataset whose file order happens to carry structure, such as a report sorted by intensity, will show horizontal banding that is a property of the sort and not of the experiment.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw: Protein, Phospho or Phosphoprotein. Only the levels actually uploaded are offered; Phosphoprotein appears once a collapse has been run.	Protein	page
Toggle IDs	Adds the original sample identifier to each column label, as condition_n (id).	on	page
Width, Height, DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page. Width and height are held between 1 and 60 cm, the resolution between 72 and 600.	10 cm × 10 cm, 300	page
Download format	PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Title	Shown or hidden, and its text, size, weight, colour and position. The title is Missing Values Heatmap at every data level.	shown	appearance
X axis, Y axis	The text and styling of Sample and Protein Numbers, and of the column labels, which are printed vertically at size 7.	plot's own	appearance
Sample names	One row per sample, to replace the condition_n label with a name of your own. Empty means the generated label.	generated	appearance
Frame and grid	Which of the four spines are drawn, and the grid behind the cells.	four spines; no grid	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Background	Theme, paper colour and transparency. The paper is only visible in the margins: the matrix itself covers the axes completely.	Default, white, opaque	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here. There is no colour card here: the palette of this figure is fixed.

3 How it is computed

HOW IT IS COMPUTED

The matrix is restricted to the sample columns named in the annotation and renamed condition_n in annotation order, so the columns arrive grouped by condition. A value is missing when it is NA, when it is not a number, or when it is infinite; a zero counts as missing as well, unless the dataset was uploaded already on the $\log_2$ scale, where zero is an intensity of 1 and is a real measurement.

Each cell is then the truth value of that one test: the whole figure is the matrix $\mathbf{1}[x_{ij} = \text{NA}]$, painted with the viridis scale. A two-valued matrix takes only the two ends of

that scale, which is why the figure has exactly two colours and no colour bar is drawn. No row or column is sorted, clustered or aggregated. Every feature with a value in at least one selected sample is one row of the image; a feature with no value in any of them is left out, as on the other completeness pages.

> **MORE FEATURES THAN PIXELS**
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> *Every* quantified feature is a row, and the image has a finite number of pixel rows: height in centimetres, divided by 2.54, times the DPI. At the default 10 cm and 300 dpi that is about 1180 rows of pixels, so a dataset of 8000 proteins cannot give each feature its own row and neighbouring features are merged into one line of the image. Isolated single missing values therefore thin out or vanish at large feature counts, while stripes and blocks survive. Raise the height and the DPI before concluding that a region is clean.

Edge cases

- A sample with no values at all is a solid yellow column; a feature measured everywhere is a solid dark row.
- A dataset with nothing missing is a solid dark rectangle, and one with nothing measured a solid yellow one. Neither draws a legend to say which it is: the two states are distinguished only by colour, so a uniform figure has to be read against the coverage plot or the Tables page.
- The y axis is labelled Protein Numbers at every data level, including phosphosite and phosphoprotein. The Y axis card in the appearance panel replaces the text.
- Filtering and imputation do not reach this page. It always draws the matrix as uploaded, which is the only matrix on which the question makes sense — an imputed matrix has no gaps left to show.

Interpreting the pattern

A column with many missing entries identifies a sample to investigate; a condition-aligned block identifies structured missingness worth checking against the design. Neither pattern establishes its cause or justifies automatic exclusion. Horizontal bands can reflect low detectability or genuinely restricted occurrence. Check feature identities, coverage and the original export before choosing a filter or imputation assumption.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = missing_value_heatmap(data, meta, id=False, data_scale="log2")
```

R

```
fig <- missing_value_heatmap(data, meta, id = FALSE, data_scale = "log2")
```

id=False drops the bracketed sample identifier from the column labels, leaving condition_n; the page has **Toggle IDs** on by default, so the figure above is the id=True labelling. The one trap is the figure size: width and height are inches here, not the centimetres of the download controls, and they default to 20 × 10 — so the number of feature rows the image can resolve, the thing this page depends on most, is set by a different unit in the notebook than on the page.

5 Exporting

The download reproduces the plot at the width, height and resolution set on the page, which is why those three controls do not change the figure on screen. Height and DPI matter more here than on any other page, because they decide how many features the image can resolve; a heatmap meant to be read at the level of individual cells needs both raised well above their defaults. PDF keeps the figure scalable, but the matrix inside it is still drawn at the chosen resolution.

Behind this page

Function	What it produces
missing_value_heatmap	The heatmap, with the sample renaming and the two-colour matrix
missing_sample_table	The per-sample counts behind a yellow column, as numbers